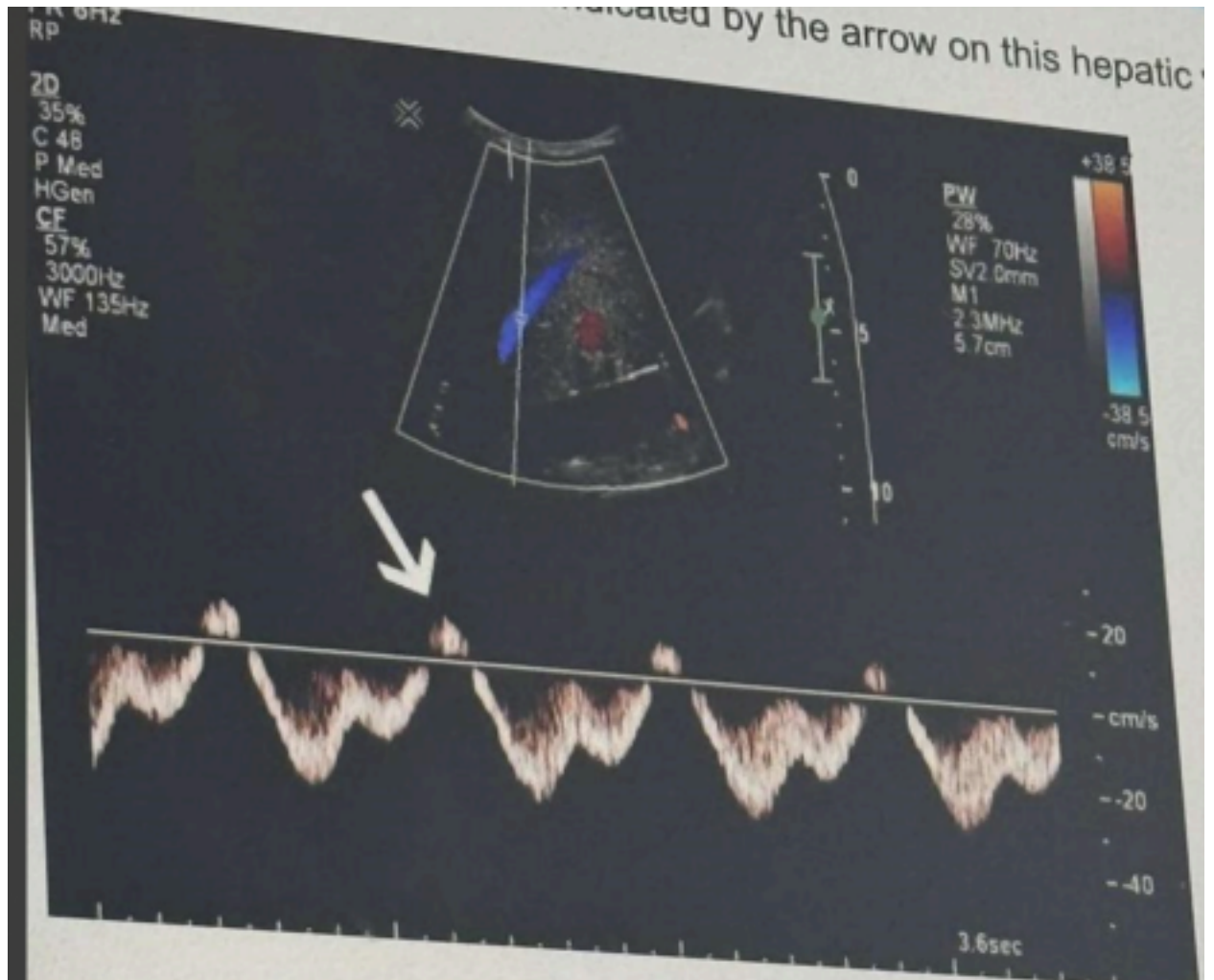
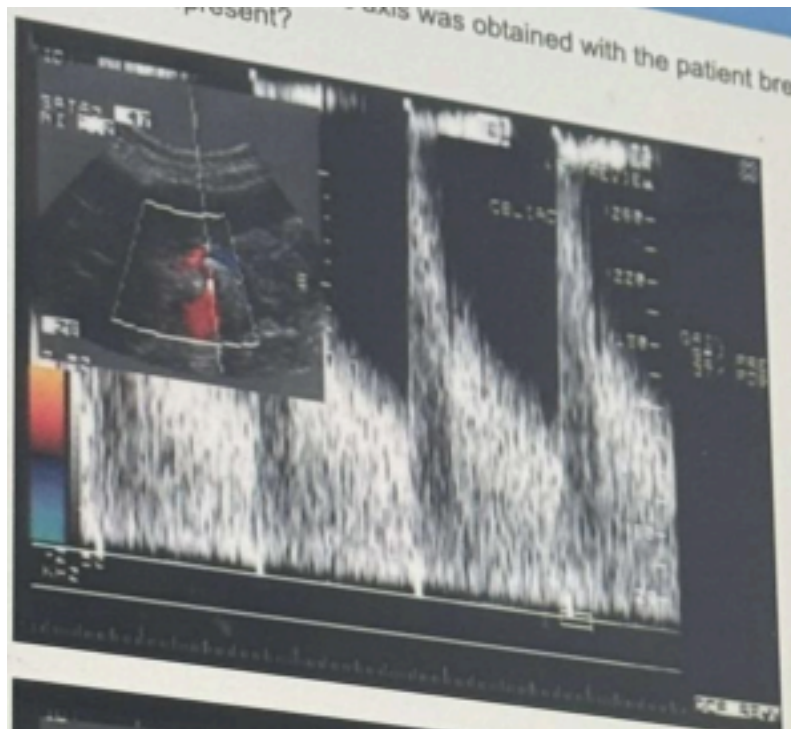


1. When performing an ankle-brachial index (ABI), which waveform characteristic is associated with medial wall calcification?
 - a. 0.60; monophasic
 - b. 0.90; triphasic
 - c. 1.15; triphasic
 - d. 1.35 monophasic
2. A vertebralbasilar duplex exam is clearly indicated due to the presence of which nonhemispheric sign?
 - a. Hemiparesis
 - b. Dysphasia
 - c. Amaurosis fugax
 - d. Syncope
3. Which vessel is most likely to be found as a singular vessel in the lower extremity venous system?
 - a. Peroneal
 - b. Anterior tibial
 - c. Posterior tibial
 - d. Common femoral
4. Which condition is associated with blue toe syndrome?
 - a. Diabetic neuropathy
 - b. Intrinsic compression
 - c. Aortoiliac occlusive disease
 - d. Inferior vena cava thrombosis
5. Which duplex exam follow-up timeframe is recommended for post-carotid endarterectomy?
 - a. Prior to 1 week
 - b. 1 to 3 weeks
 - c. 1 to 3 months
 - d. 6 months
6. Tumor invasion into the inferior vena cava is most commonly associated with which condition?
 - a. Renal cell carcinoma
 - b. Ovarian carcinoma
 - c. Hepatocellular carcinoma
 - d. Budd-chiari syndrome
7. Which cardiac cycle event is indicated by the arrow on the hepatic vein spectral doppler image?



- Right atrium contracting
 - Right atrium filling in systole
 - Right atrium filling in diastole
 - Right atrium overfilling before the tricuspid valve opens
8. The top image of the celiac axis was obtained with the patient breathing normally. The lower image was obtained...most likely represent?



- a. A normal finding
 - b. Median arcuate syndrome
 - c. Celiac axis stenosis
 - d. Leriche syndrome
9. Which finding is considered normal after a carotid endarterectomy?
- a. Increased size of bulb
 - b. Decreased size of bulb
 - c. Increased size of the proximal external carotid artery
 - d. Decreased size of the proximal internal carotid artery
10. What is considered a normal ankle-brachial index (ABI) value?
- a. Less than 0.5

- b. 0.5-0.8
- c. 0.9-1.3
- d. 1.5-2.0

11. What condition is likely when a patient presents with intermittent pallor followed by cyanosis of the ha...

- a. Peripheral emboli
- b. Atherosclerosis disease
- c. Incomplete palmar arch
- d. Raynaud disease

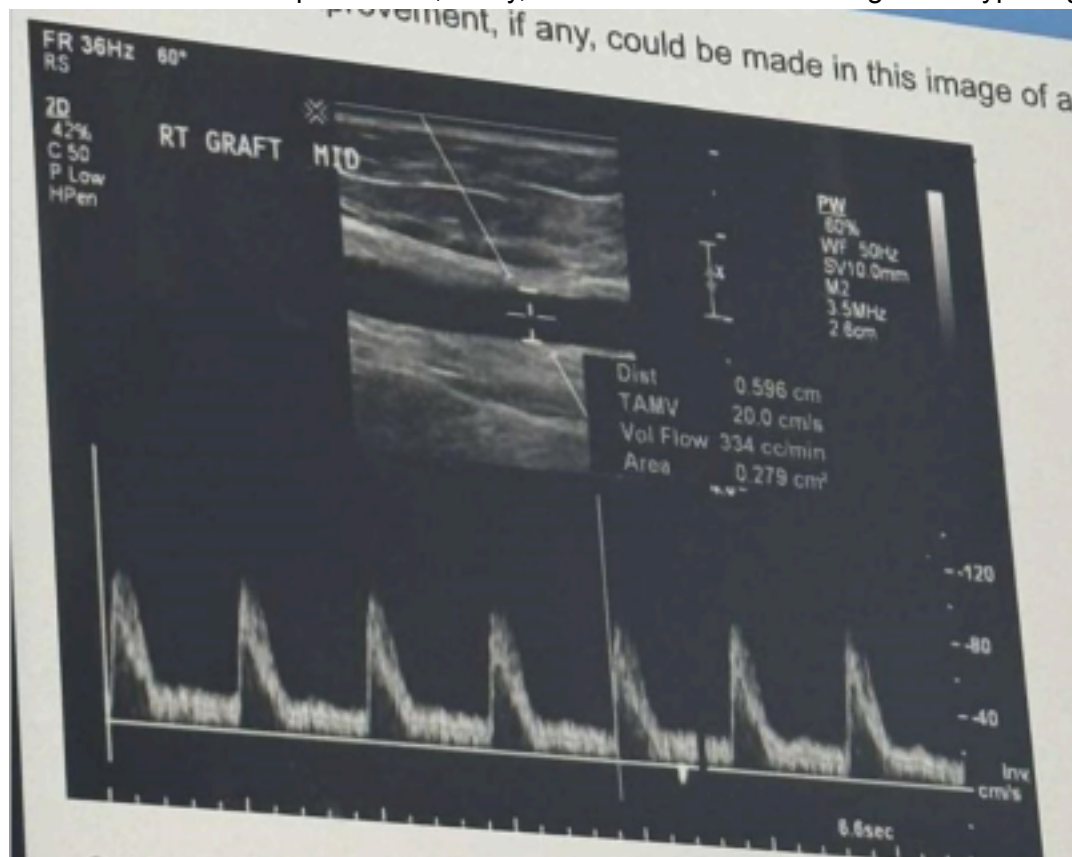
12. What is the normal direction of flow in the portal and hepatic veins?

- a. Both are hepatopetal
- b. Both are hepatofugal
- c. Portal vein flow is hepatopetal; hepatic vein flow is hepatofugal
- d. Portal vein flow is hepatofugal; hepatic vein flow is hepatopetal

13. Which condition is most likely to cause decreased femoral pulses and an abdominal bruit in an 18-year-old hypertension m...

- a. Fibromuscular dysplasia
- b. Arteriovenous fistula
- c. Neurovascular compression syndrome
- d. Coarctation of the aorta

14. Which technical improvement, if any, could be made in this image of a bypass graft?



a.

Make no change because image is technically satisfactory

- b. Decrease sample volume

- c. Adjust doppler spectrum cursors
- d. Adjust diameter measurement

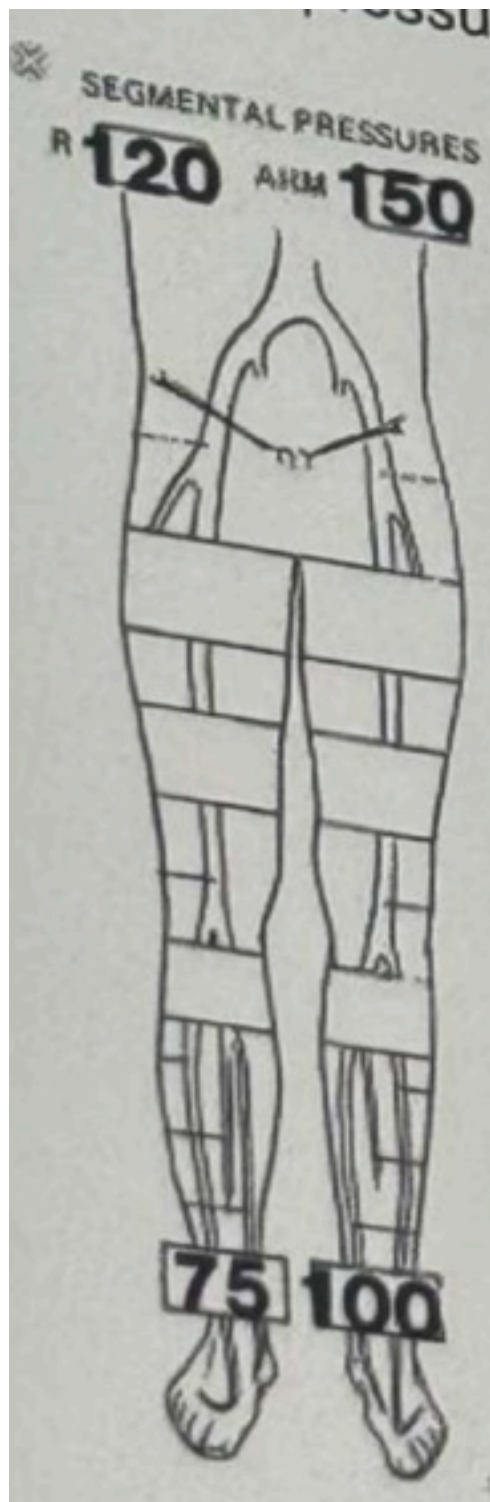
15. Obstruction of which vein is inferred with the unilateral finding of a continuous doppler waveform in the subclavian vein?

- a. Internal jugular
- b. Brachiocephalic
- c. Azygos
- d. Superior vena cava

16. Which finding is most consistent with a thumb photoplethysmographic (PPG) waveform that shows no change with radial artery compression?

- a. Patent radial artery
- b. Occluded ulnar artery
- c. Incomplete palmar arch
- d. Complete palmar arch

17. Using pressures given in this image, what is the ankle-brachial index for the right leg?

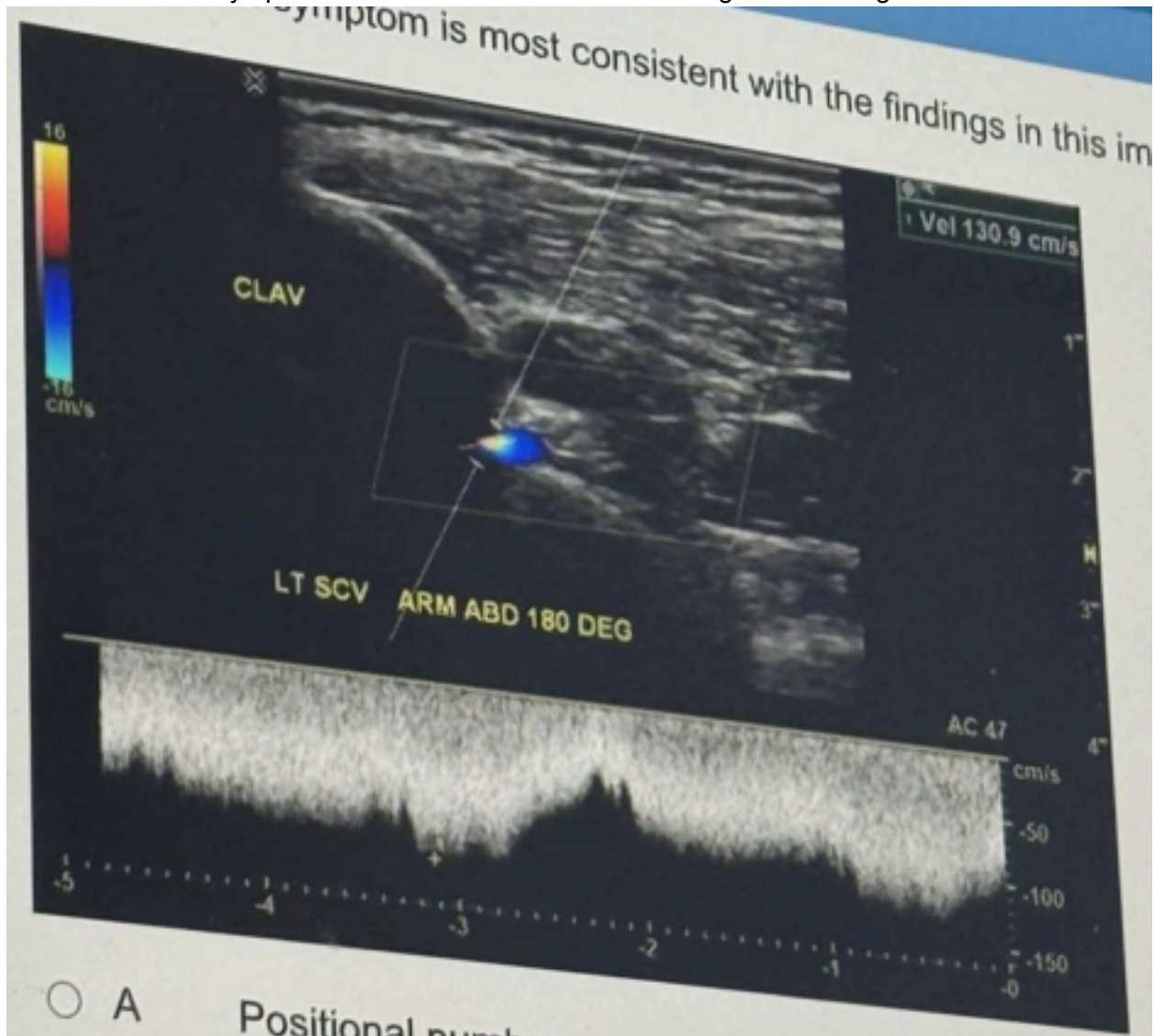


- a. 0.50
- b. 0.63
- c. 0.67
- d. 0.83

18. Which sonographic orientation is used to assess the lower extremity veins for complete compressibility?

- a. Coronal
- b. Sagittal
- c. Transverse
- d. Oblique

19. Which clinical symptom is most consistent with the findings in this image?

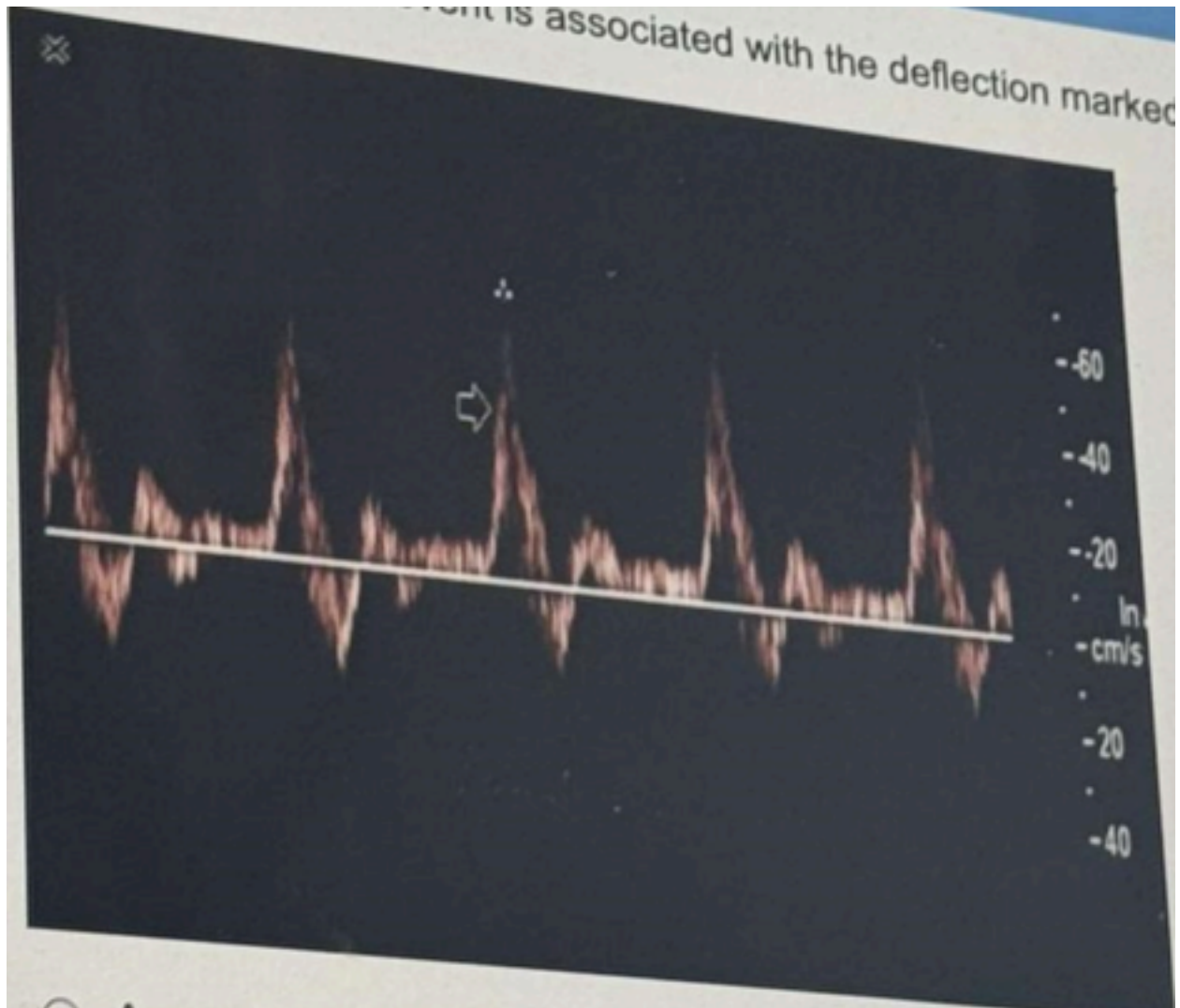


- a. Positional numbness
 - b. Positional swelling
 - c. Digital ischemia
 - d. Digital blanching
20. Doppler signal that has a continuous forward flow throughout the cardiac cycle would normally be observed in which artery under resting or fasting conditions?
- a. Hypogastric
 - b. Deep femoral
 - c. Celiac axis
 - d. Superior mesenteric
21. Which artery is indicated by the arrow in this image?

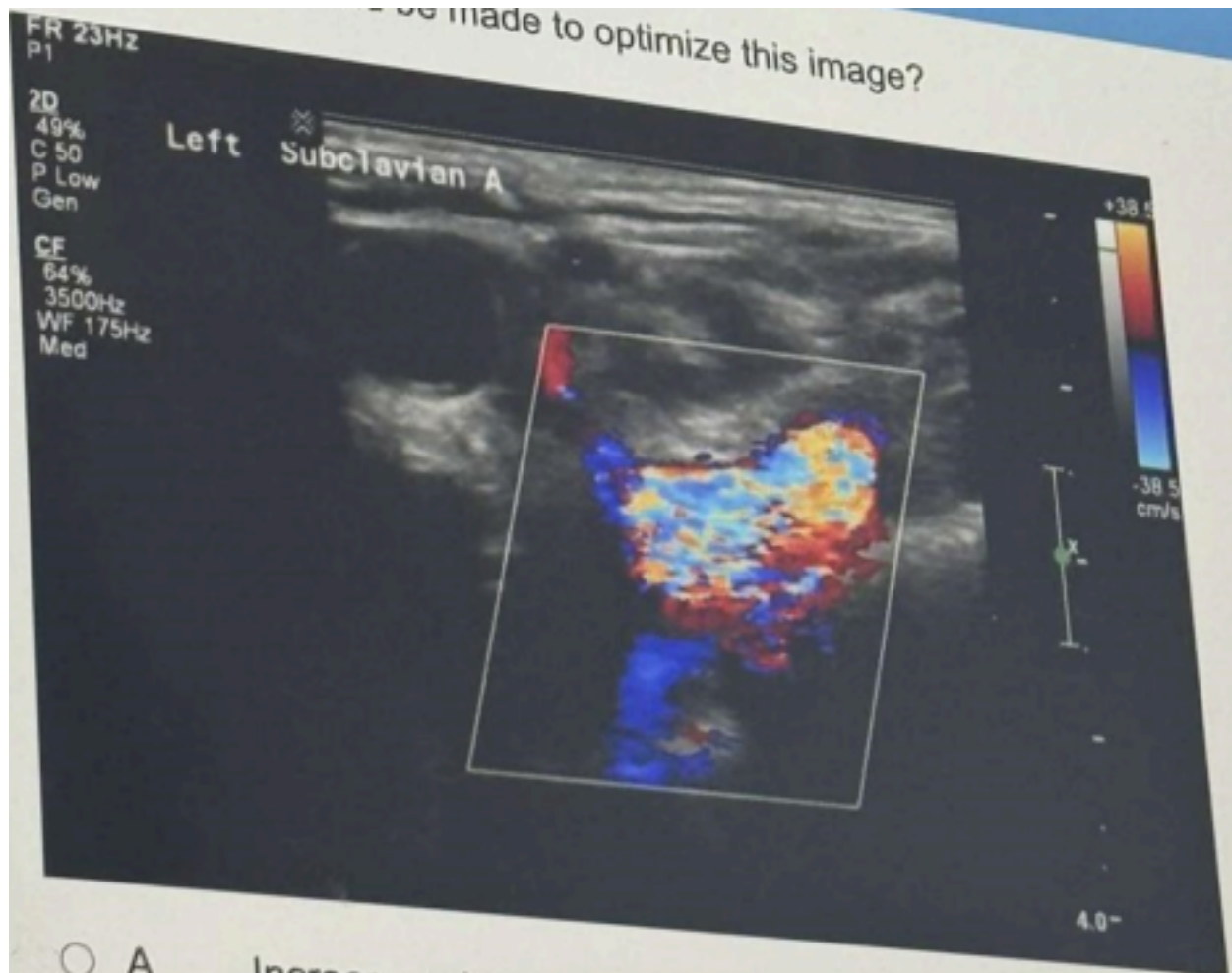


- a. Common carotid bifurcation
- b. Proximal internal carotid
- c. Distal common carotid
- d. Proximal external carotid

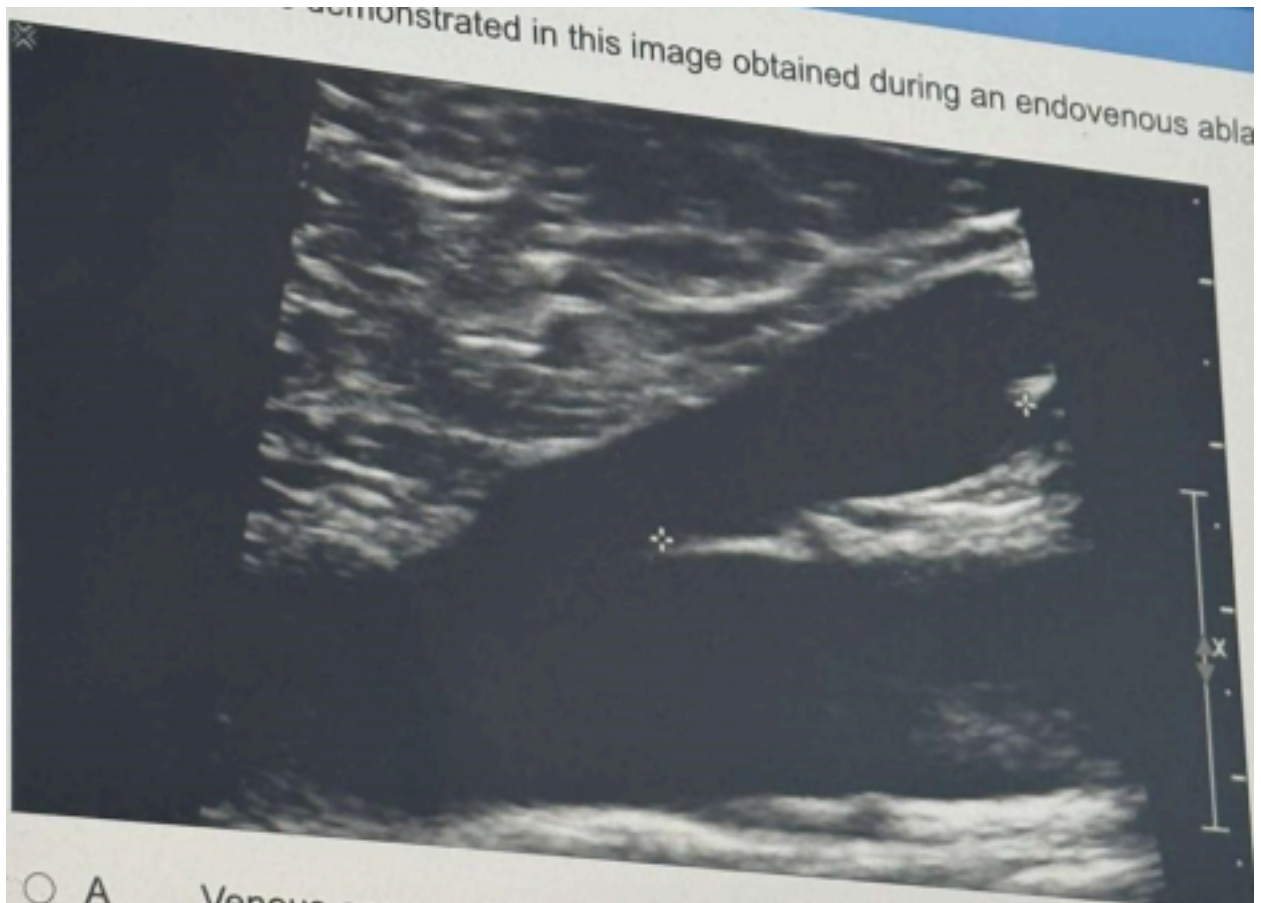
22. Which cardiac cycle event is associated with the deflection marked by the arrow on this waveform?



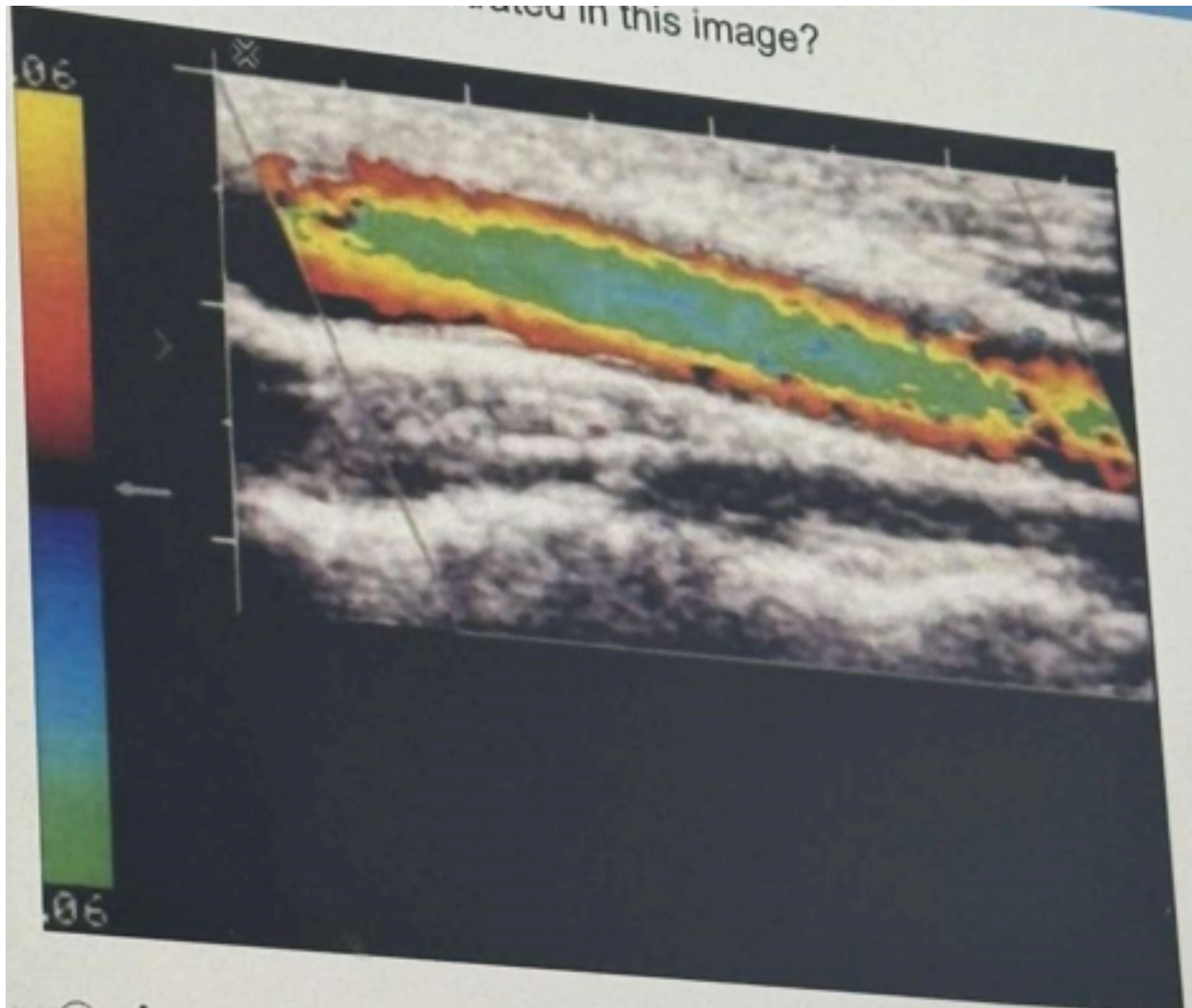
- a. Elastic recoil of the vessel wall
 - b. Ventricular systole
 - c. Atrial systole
 - d. Peripheral resistance
23. Which change should be made to optimize this image?



- a. Increase color scale
 - b. Decrease color scale
 - c. Increase color gain
 - d. Decrease overall gain
24. In addition to atherosclerosis, what is another cause of flow-reducing stenoses of the main renal arteries?
- a. Parenchymal disease
 - b. Fibromuscular dysplasia
 - c. Glomerulonephritis
 - d. Tubular necrosis
25. Which situation is demonstrated in this image obtained during an endovenous ablation?



- a. Venous access site
 - b. Catheter placement
 - c. Vessel depth measurement
 - d. Tumescence anesthesia injection
26. Which artifact is demonstrated in this image?



- a. Color flash
- b. Aliasing
- c. Side-lobe
- d. Grating lobe

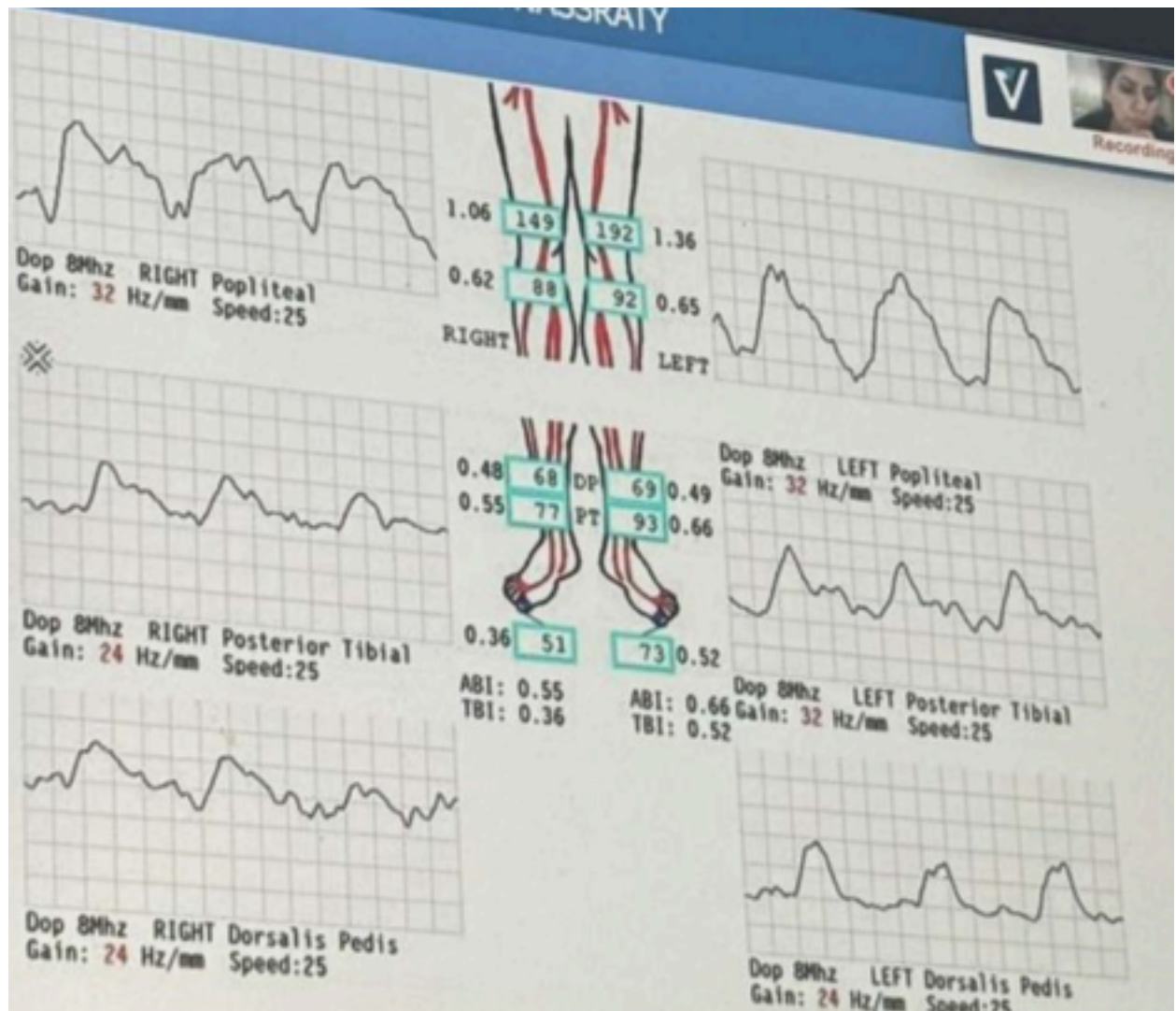
27. What does continuous flow in the common femoral vein suggest?

- a. Chronic obstructive pulmonary disease
- b. Congestive heart failure
- c. Distal obstruction
- d. Proximal obstruction

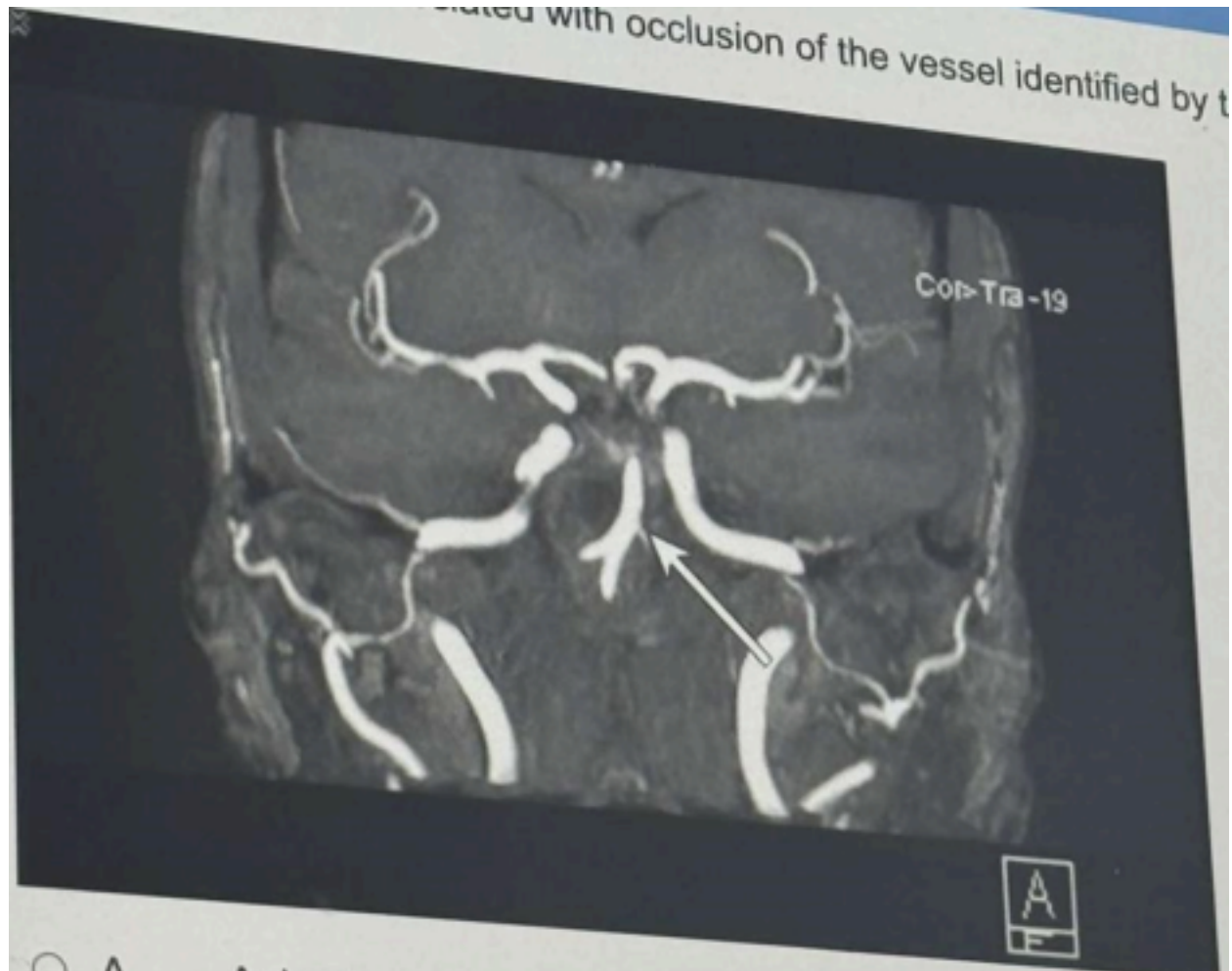
28. What technique is optimal for evaluation of palmar arch patency?

- a. 1st and 5th digit pressures during brachial and axillary artery compression
- b. 1st and 5th digit waveforms during radial and ulnar artery compression
- c. Radial and ulnar artery pressures during 1st and 5th digit compression
- d. Radial and ulnar artery waveforms during brachial and axillary artery compression

29. No question?



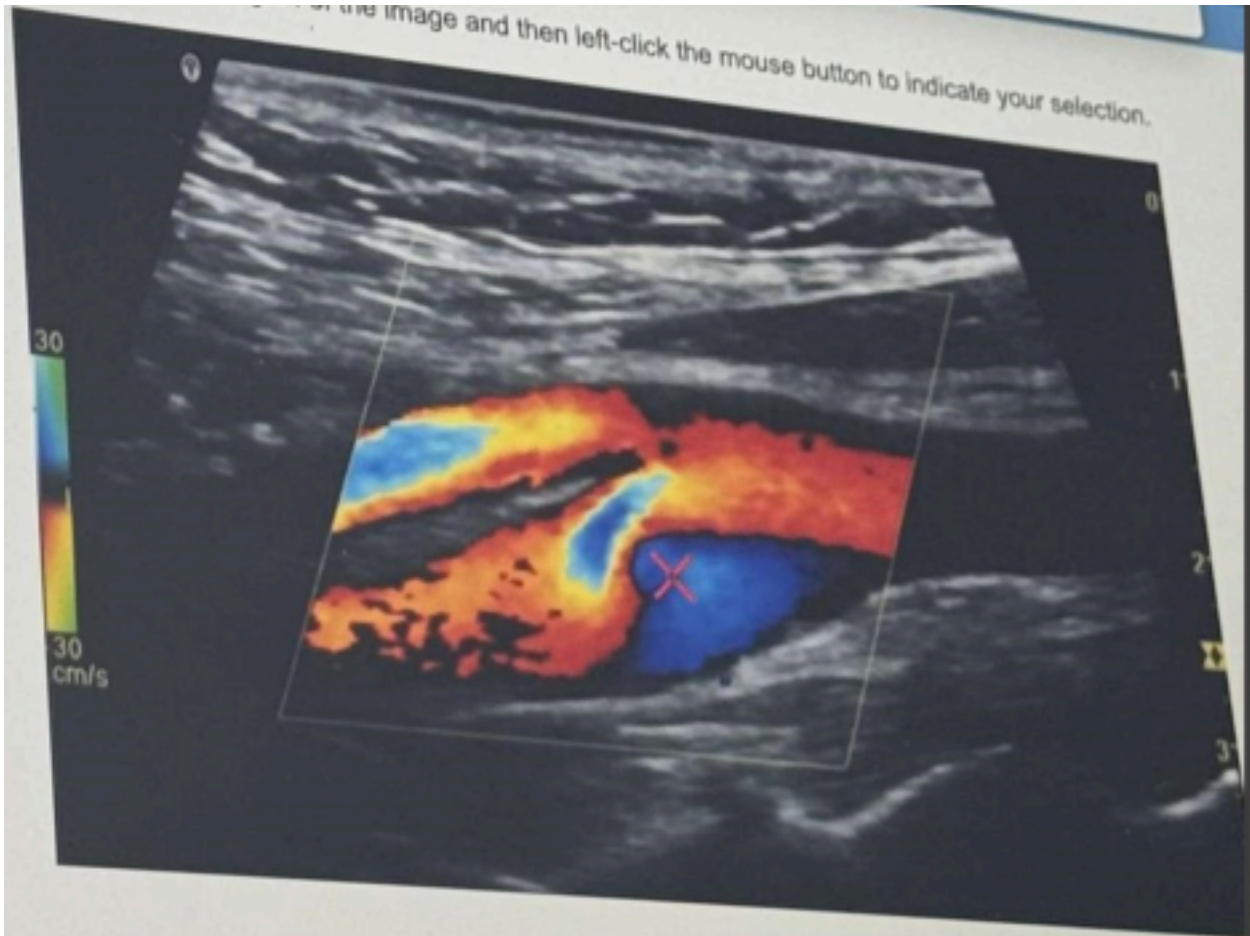
- a. Gangrene
 - b. Claudication
 - c. Ischemic rest pain
 - d. Non-healing ulcer
30. Which symptom is associated with occlusion of the vessel identified by the arrow on this image?



- a. Aphasia
- b. Paresthesia
- c. Diplopia
- d. Hemiparesis

31. Identify the area that demonstrates flow reversal.

Using your mouse, place the cursor on the appropriate region of the image and then left-click the mouse button to indicate your selection.



32. What is the most likely cause of a carotid artery stent obstruction in the absence of stent fracture or deformation?

- a. Intraplaque hemorrhage
- b. Intimal hyperplasia
- c. Arterial thrombosis
- d. Dissection

33. When the cardiac cycle does reversal of flow occur in a normal lower extremity arterial waveform?

- a. Peak systole
- b. End diastole
- c. Late systole/early diastole
- d. Late diastole/early systole

34. Which measurement is the lowest peak systolic velocity associated with a superior mesenteric artery stenosis of 70% in a fasting patient?

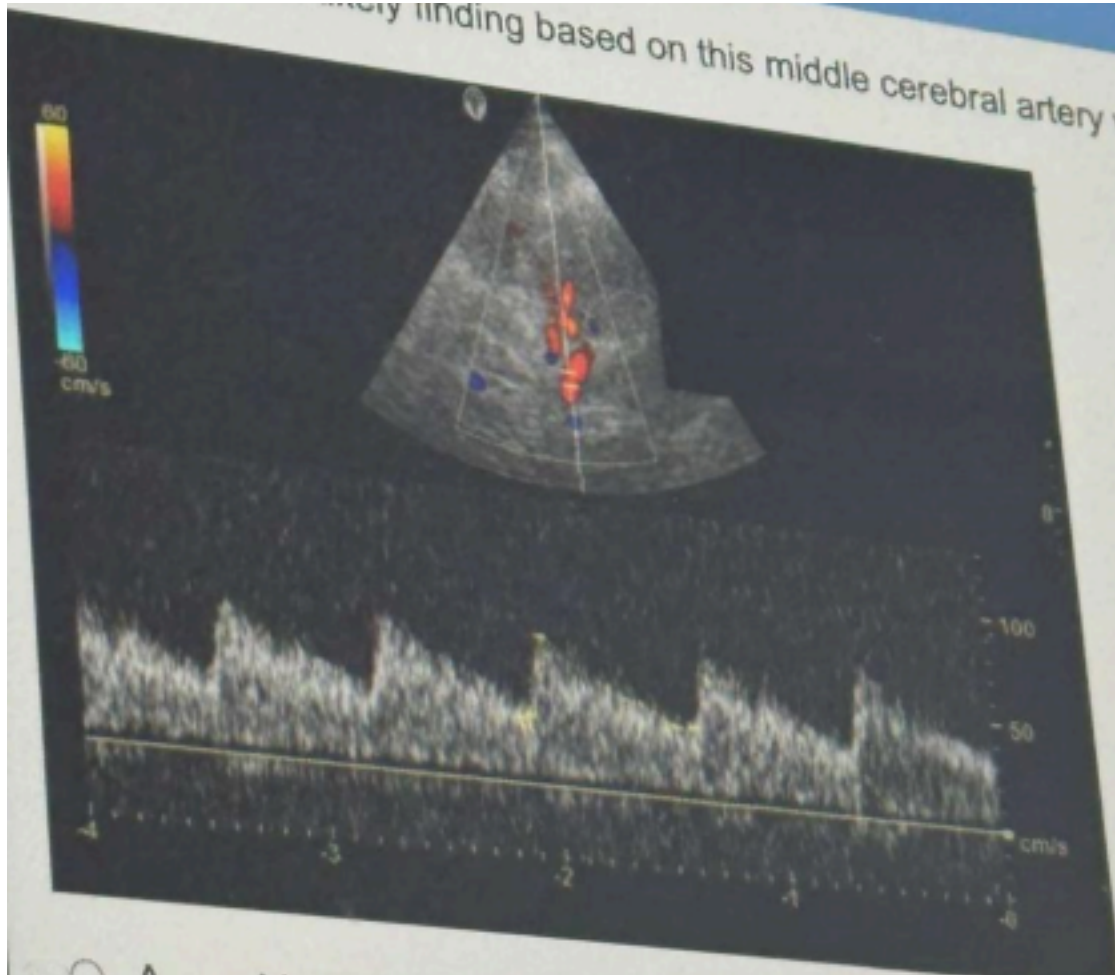
- a. 150 cm/s
- b. 200 cm/s ---
- c. 275 cm/s
- d. 325 cm/s

35. Which condition may be diagnosed based on a venous refill time of 10 seconds without a tourniquet that changes to 27 seconds when a tourniquet is applied above the knee? a.

- Incompetence of the popliteal vein
- b. Incompetence of the femoral vein

- c. Incompetence of the great saphenous vein
- d. Valvular insufficiency of the small saphenous vein

36. Which is the most likely finding based on this middle cerebral artery waveform?



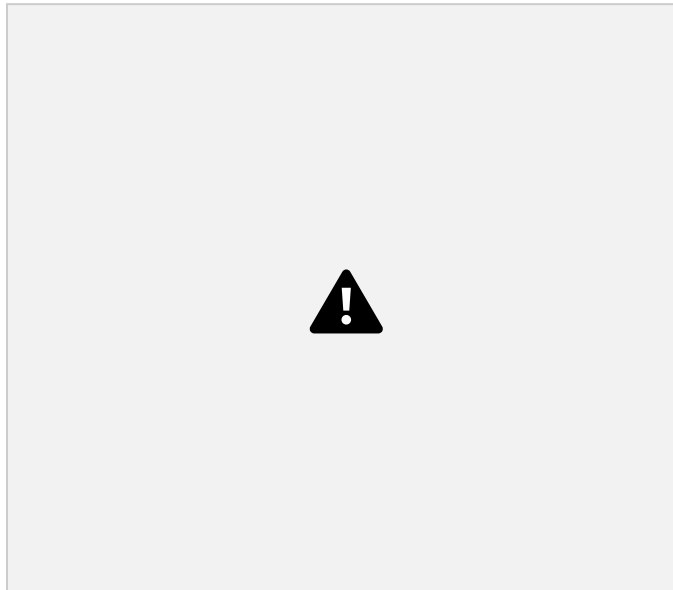
- a. Vasospasm
 - b. Stenosis
 - c. Normal flow
 - d. Distal occlusion
37. A right eye visual disturbance is most commonly caused by stenosis of which artery?
- a. Right internal carotid
 - b. Right external carotid
 - c. Left external carotid
 - d. Left internal carotid
38. Which maneuver helps in the diagnosis of an iatrogenic arteriovenous fistula?
- a. Valsalva
 - b. Distal augmentation
 - c. Manual occlusion of proximal vein
 - d. Repeated ankle flexion and extension
39. What describes the waveform of a normal vertebral artery?
- a. Pandiastolic flow reversal
 - b. Multiphasic diastolic flow

- c. Absence of diastolic flow
- d. Forward flow through diastole

40. What occurs during expiration?

- a. Diaphragm ascends
- b. Intrathoracic pressure decreases
- c. Intra-abdominal pressure increases
- d. Venous flow from the upper extremities increases

41. Which waveform represents the turbulence associated with the stenosis in this image?



- a. A
- b. B
- c. C**
- d. D

42. What is the most common site for embolization from a subclavian artery aneurysm?

- a. Digit
- b. Forearm
- c. Elbow
- d. upper arm

43. Which condition is characterized by blood flow separating the layers of the artery wall?

- a. Ectasia
- b. Dissection
- c. Adventitial cystic disease
- d. medial calcinosis

44. How is an image affected when the color Doppler scale is set too low?

- a. produces aliasing
- b. aliasing can be eliminated
- c. color bleed is decreased
- d. no color is seen within the vessel

45. Occlusion of which artery would cause flow reversal to the right vertebral artery?

- a. Innominate
- b. left vertebral
- c. left subclavian
- d. right common carotid

46. What type of aneurysm can be treated with thrombin injection?

- a. Fusiform
- b. Saccular
- c. False
- d. Mycotic

47. Which systolic pressure is measured at each cuff level?

- a. artery segment proximal to cuff
- b. artery segment under cuff
- c. artery segment distal to cuff
- d. artery segment proximal and distal to cuff

48. What effect does deep exhalation have on venous flow?

- a. flow in the upper and lower extremities increases
- b. flow in the upper and lower extremities decreases
- c. flow in the upper extremities increases; flow in the lower extremities decreases
- d. flow in the upper extremities decreases; flow in the lower extremities increases

49. What is revealed in the spectral analysis of the vein in the image below?



- a. Venous outflow obstruction
 - b. Venous insufficiency
 - c. Venous Competency
 - d. Normal venous flow
- 50. Which description best characterizes normal flow in the cortical renal arteries?
 - a. Low velocity, High resistive
 - b. low velocity, low resistive
 - c. High velocity, High resistive
 - d. high velocity, low resistive
- 51. Which condition is most likely associated with the finding demonstrated in this image?



- a. carotid body tumor
 - b. fibromuscular dysplasia
 - c. carotid pseudoaneurysm
 - d. carotid dissection
52. Which spectral Doppler feature is consistent within an occluded arteriovenous fistula (AVF)?
- a. Inflow artery, proximal to the AVF anastomosis, showing low resistive pattern
 - b. outflow vein showing no arterial pulsations
 - c. inflow artery, distal to the AVF anastomosis, showing reversal of flow
 - d. outflow vein showing no respiratory variations
53. Identify the second branch of the abdominal aorta.
- Using your mouse, place the cursor on the appropriate region of the image and then left click the mouse button to indicate your selection



54. The Doppler spectral waveform from the superior mesenteric artery in a normal fasting subject demonstrates which Doppler pattern?
- a. High diastolic velocity
 - b. Low diastolic velocity
 - c. monophasic pattern
 - d. increased Peak systolic velocity
55. Which arterial lesion would be most amenable to percutaneous transluminal angioplasty?
- a. 20 cm occlusion of The Superficial femoral artery
 - b. focal 90% iliac artery stenosis
 - c. 4 cm occlusion of the posterior tibial artery
 - d. popliteal artery aneurysm

56. Which arterial anatomic location corresponds with a pressure gradient between the brachial artery and the high thigh?

- a. Aorta
- b. radial artery
- c. popliteal artery
- d. posterior tibial artery

57. Which structure should be targeted during ultrasound guided compression of a pseudoaneurysm?

- a. Pseudoaneurysm neck
- b. Pseudoaneurysm sac
- c. artery adjacent to the pseudoaneurysm
- d. vein adjacent to the pseudoaneurysm

58. What waveform characteristics would be expected on spectral Doppler analysis of a normal ophthalmic artery?

- a. Retrograde; sharp systolic upstroke
- b. Antegrade; sharp systolic upstroke
- c. Retrograde; slowed systolic upstroke
- d. Antegrade; slowed systolic upstroke

59. What is the duplex finding that most accurately determines internal Carotid artery occlusion?

- a. Absence of color in the internal carotid artery
- b. decreased ipsilateral common artery flow velocity
- c. absence of Doppler signal in the internal Carotid artery
- d. increased diastolic flow in the common carotid artery

60. Which renal artery condition is most likely represented in this image?



- a. Occlusion
 - b. Dissection
 - c. Aneurysm
 - d. fibromuscular dysplasia
61. Which vessel is indicated by the arrow in this image?



- a. Left renal vein
- b. left renal artery
- c. right renal vein
- d. right renal artery

62. Identify the vessel that should demonstrate a high resistance signal in a fasting state. Using your mouse, place the cursor on the appropriate region of the image and then left-click the mouse button to indicate your selection.



63. What clinical finding is associated with lower extremity Venous insufficiency?

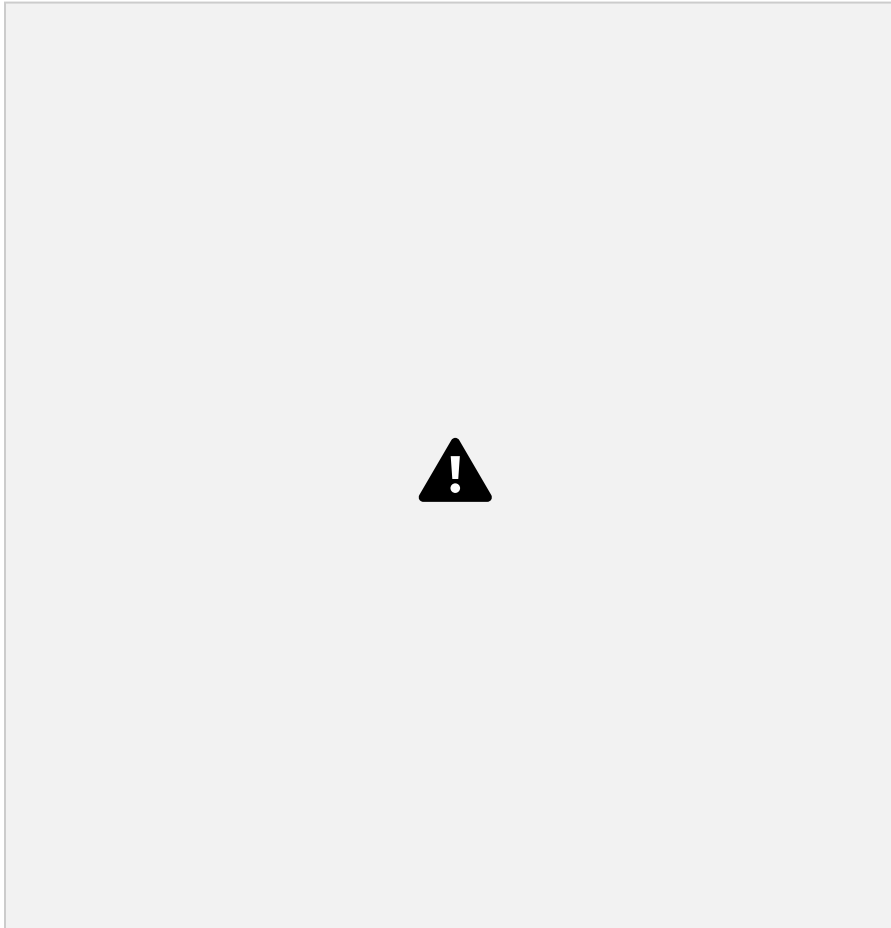
- a. Hyperpigmentation
- b. Dependent rubor
- c. Pulselessness
- d. hair loss

64. Which finding will be visualized when a patient presents with an arteriovenous fistula following trauma?

- a. Increased proximal venous flow

- b. decreased proximal arterial flow
- c. increased distal arterial flow
- d. decreased proximal Venous flow

65. Which finding is represented by the finely echogenic component within the vein indicated by the V on this image?



- a. Thrombosis
 - b. Rouleaux flow
 - c. elevated velocity
 - d. reverberation artifact
66. Identify the structure that causes venous insufficiency.
- Using your mouse, place the cursor on the appropriate region of the image and then left-click the mouse button to indicate your selection.



67. Which sonographic finding is associated with this lower extremity waveform?



- a. distal obstruction
- b. Proximal obstruction
- c. collateral flow
- d. venous insufficiency

68. An ultrasound examination was performed one week after the thermal ablation of the great saphenous vein. Which ... Indicated by the arrow in this image?



- a. Normal flow in The Superficial epigastric vein
- b. normal flow in The Superficial epigastric artery
- c. reversed flow in The Superficial epigastric vein
- d. reversed flow in The Superficial epigastric artery

69. What is the correct setting of the sample volume when measuring flow volumes in a dialysis access?

- a. Expanded to the entire vessel diameter
- b. narrowed down to 1 mm
- c. adjacent to anterior vessel wall
- d. adjacent to posterior vessel wall

70. Identify the structure that causes venous insufficiency.

Using your mouse, place the cursor on the appropriate region of the image and then

left-click the mouse button to indicate your selection.



71. What is suggested when the arterial signal persists during performance of an ankle-brachial index with the ankle cuff maximally inflated?

- a. triphasic waveform
- b. arterial calcification
- c. pseudoaneurysm
- d. arterial stenosis

72. Which patient position is typically used during endovenous ablation of the small saphenous vein?

- a. Prone
- b. Supine
- c. Standing
- d. Seated

73. Where does the blood from the soleal venous sinuses drain?

- a. Deep femoral vein
- b. popliteal vein
- c. femoral vein
- d. peroneal vein

PTV

74. A Doppler signal that has a continuous forward flow through the cardiac cycle would normally be observed in which artery under res....?

- a. Hypogastric
- b. deep femoral
- c. Celiac axis
- d. Superior mesenteric

75. Which reason best explains why the spectral waveforms of the proximal abdominal aorta have different characteristics than those of the distal abdominal aorta?

- a. Proximal aorta is larger than distally aorta also true
- b. proximal aorta supplies the visceral arteries
- c. distal aorta supplies essential organs
- d. distal aorta has more branches

76. What level of disease is expected if a significant pressure gradient is present from the low thigh to the calf?

- a. Iliofemoral
- b. Femoropopliteal
- c. Tibioperoneal
- d. Aortoiliac

77. Which criteria is most important for identification of the upper extremity deep veins? a. adjacent artery

- b. size of vessel
- c. depth of vessel
- d. course of vessel

78. Identify the structure that causes venous insufficiency.

Using your mouse, place the cursor on the appropriate region of the image and then left-click the mouse button to indicate your selection.



79. Which measurement is most consistent with proximal renal artery stenosis?
- a. Renal aortic ratio (RAR) of 2.8
 - b. parenchymal resistive index of 0.80
 - c. parenchymal acceleration time of 0.3 seconds
 - d. renal artery Peak systolic velocity of 170 cm/s
80. Which abnormality is associated with intermittent buttock and thigh claudication?
- a. May-thurner syndrome
 - b. Buerger disease
 - c. aortoiliac occlusive disease
 - d. superficial femoral artery stenosis
81. A young adult male is seen in the emergency department after a small-caliber gunshot

wound of the left pelvis. a palpable left femoral pulse rules out which condition? a.

Arteriovenous fistula

- b. Venous laceration
- c. iliac vein occlusion
- d. iliac artery occlusion

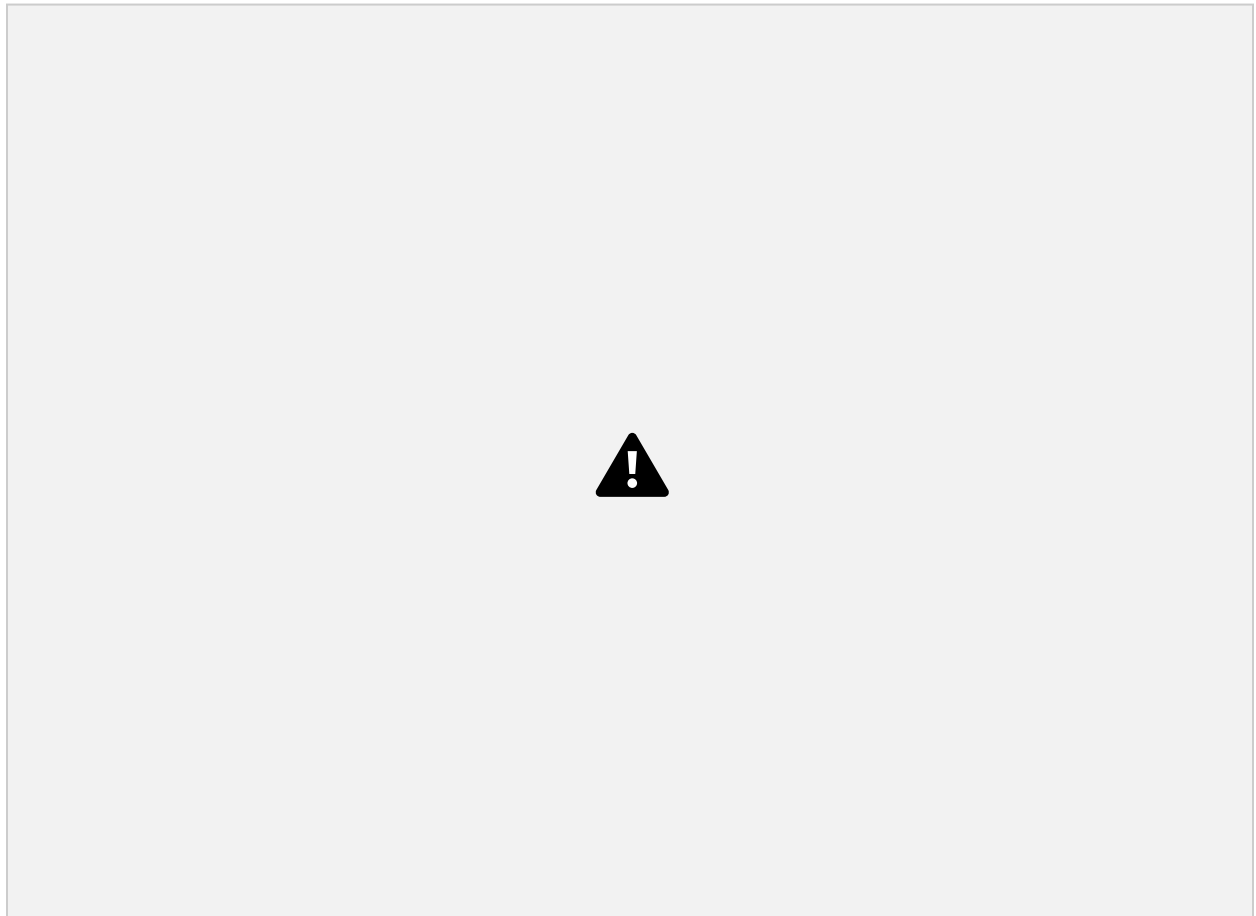
82. Which technique should we use when evaluating the central veins?

- a. Compression
- b. Doppler analysis
- c. apply tourniquet
- d. Photoplethysmography

83. Which value is associated with normal valve closure time?

- a. 0.4 sec
- b. 0.6 sec
- c. 1.0 sec
- d. 1.2 sec

84. Which finding is demonstrated in this image?



- a. Spectral broadening
- b. pulse repetition frequency set too high
- c. mirror image artifact
- d. aberration of normal anatomy

85. Identify the artery most likely compressed by the median arcuate ligament. Using your mouse place the cursor on the appropriate region of the image and then left-click the mouse button to indicate your selection. Celiac trunk



86. Spectral analysis of the right common carotid artery demonstrates a low velocity, dampened waveform. pathology in the common carotid artery is not seen. what is the most likely cause? a. Intracranial internal carotid artery occlusion
b. subclavian steal
c. contralateral common carotid artery occlusion
d. innominate artery occlusion

87. A patient walked on a treadmill at 2 mph and at a 10% incline for 5 minutes. The post-exercise Ankle pressures improved when compared to the resting ankle pressures.

however the patient experienced calf, thigh, and buttock pain when walking. What is the preliminary interpretation of the test?

- a. Single-segment arterial occlusive disease is the cause of the patient's symptoms
- b. two-segment arterial occlusive disease is the cause of the patient's symptoms
- c. multi-segment arterial occlusive disease is the cause of the patient's symptoms
- d. arterial occlusive disease can be ruled out as the cause of the patient symptoms

88. What is the most likely finding in a patient presenting with patent arteriovenous fistula, ipsilateral arm and hand edema, skin discoloration, and prominent chest wall superficial veins?

- a. Occluded superficial vein
- b. steal phenomenon
- c. inflow artery occlusion
- d. central venous obstruction

89. Which finding is most consistent with a thumb photoplethysmographic (PPG) waveform that shows no change with radial artery compression?

- a. patent radial artery
- b. occluded ulnar artery
- c. incomplete palmar arch
- d. complete palmar arch

90. How does inspiration normally affect venous flow?

- a. Increases venous flow in both upper limbs and neck
- b. decreases venous flow in both upper and lower limbs
- c. decreases venous flow in both lower limbs and neck
- d. decreases venous return to heart

91. And a patient with normal arterial flow and resting ankle brachial index of 1.0, what is the expected post-exercise ankle-brachial index?

- a. 1.5
- b. 1 same or slightly increased
- c. 0.75
- d. 0.5

92. Which finding is characteristic of acute renal allograft swelling, tenderness, and infarction?

- a. Ureter compression due to peritransplant fluid collection
- b. low peak systolic velocity in the iliac artery
- c. Kinking of the main renal artery
- d. obstruction of the main renal vein

93. Which finding indicates that an arteriovenous fistula has reached maturity?

- a. Length of at least 6 millimeters for cannulation
- b. Blood flow volume of at least 600 milliliters/minute
- c. Depth of more than 6 millimeters from the skin surface
- d. systolic velocity of more than 600 centimeters/second

94. Which computer tomography (CT) finding would most likely be responsible for impeded venous outflow at the level of the iliac veins?

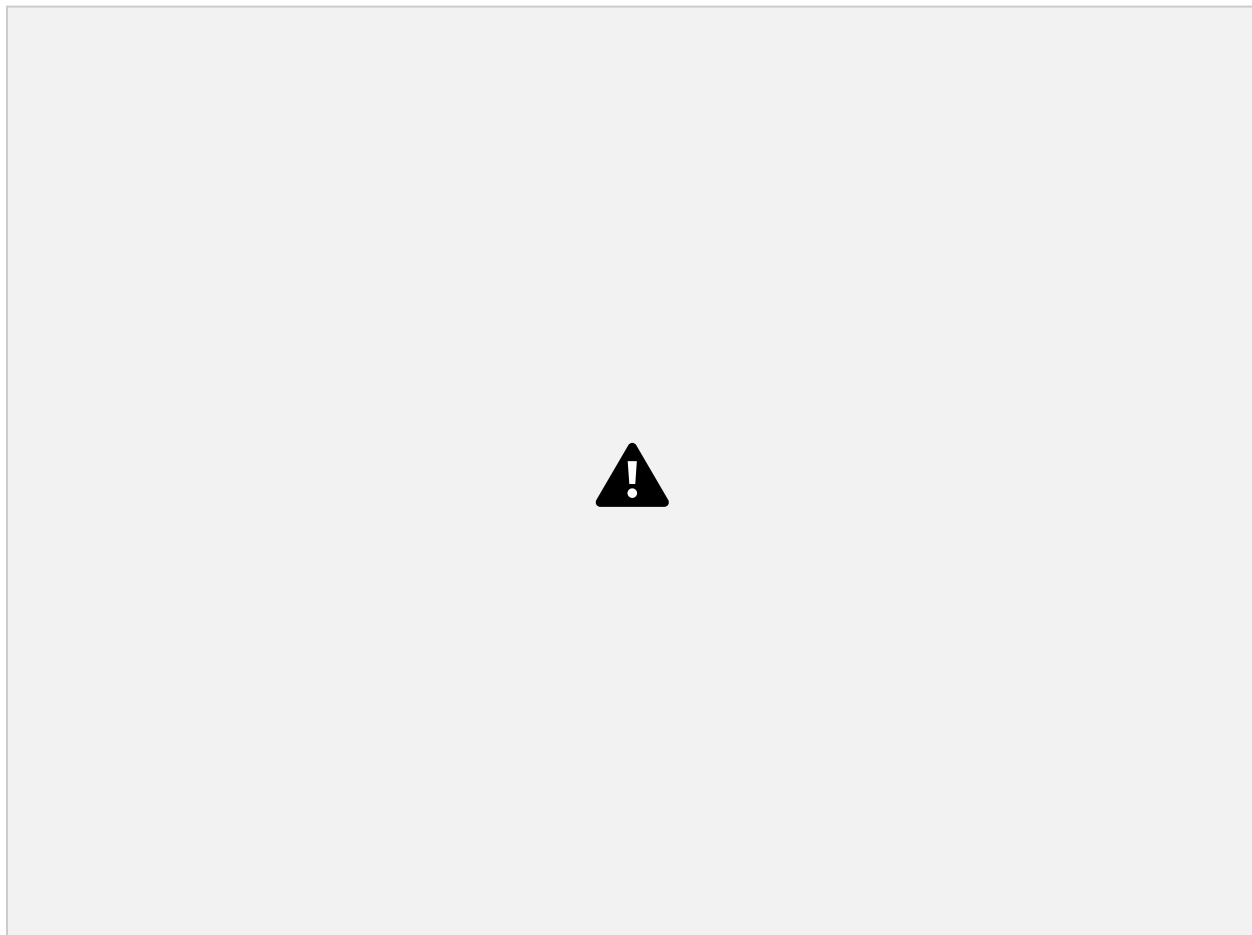
- a. Renal cyst
- b. pelvic Mass

- c. abdominal aortic aneurysm
- d. femoral vein thrombosis

95. A febrile patient with a hot, erythematous, swollen calf and ankle most likely has which condition?

- a. Acute deep vein thrombosis
- b. chronic venous insufficiency
- c. Cellulitis
- d. lymphedema

96. What structure is indicated by the arrow on this image obtained from a patient with portal hypertension?



- a. Right portal vein
- b. dilated coronary vein
- c. recanalized paraumbilical vein
- d. transjugular intrahepatic portosystemic shunt (TIPS)

97. In the presence of severe stenosis of the distal common carotid artery, which flow pattern is most likely present in the ipsilateral proximal internal carotid artery?

- a. Laminar
- b. Turbulent
- c. low resistance
- d. High Resistance

98. Which vein is responsible for venous reflux when the venous refill time normalizes after

application of a thigh tourniquet?

- a. Perforating
- b. Femoral
- c. small saphenous
- d. great saphenous

99. Which further imaging may be needed to confirm an endoleak in a post-repair abdominal aorta?

- a. Abdominal x-ray
- b. computed tomography scan
- c. Echocardiogram
- d. nuclear medicine scan

100. What is the most common cause of cerebrovascular stroke?

- a. Middle cerebral artery thromboembolism
- b. rupture of cerebral aneurysm in Circle of Willis
- c. essential hypertension
- d. brain concussion

101. What is the normal gradient between segmental pressure in a lower extremity?

- a. 20 mmHg
- b. 25 mmHg
- c. 30 mmHg
- d. 35 mmHg

102. Which statement is most consistent with arterial occlusive disease following exercise testing?

- a. Ankle pressure decreases but the pressure returns to baseline within 1 minute
- b. peripheral resistance decreases resulting in decreased blood flow
- c. peripheral resistance increases resulting in increased blood flow
- d. ankle pressure decreases by 20% or greater from the resting baseline

103. Which condition is consistent with a dampened photoplethysmogram waveform?

- a. Hyperemia
- b. complete palmar arch
- c. Raynaud Disease
- d. arteriovenous fistula

104. The top image of the Celiac axis was obtained with the patient breathing normally. The lower image was obtained when the patient took a deep breath. What ... most likely represent?



No answer

choices

Same as 8

105. Which duplex findings would suggest isolated iliac vein thrombosis?
- a. dilated incompressible common femoral vein with absent Doppler signals
 - b. compressible common femoral vein with continuous Doppler signals
 - c. dilated incompressible popliteal vein with absent Doppler signals
 - d. normal venous duplex exam with spontaneous and phasic Doppler signals

106. Identify the superior thyroid artery.

Using your mouse, place the cursor on the appropriate region of the image and then left-click the mouse button to indicate your selection.

